Chapter 2

Background Study

This chapter surveys the work on which the project builds and establishes the gap it addresses. The literature is organised into four themes: deep learning and vision foundation models for Alzheimer's diagnosis, federated learning for medical applications, client selection strategies in federated learning, and explainability in medical AI. Section 2.2 draws these threads together into the problem this work takes up.

2.1 Literature Review

2.1.1 Deep Learning and Vision Foundation Models for Alzheimer's Disease Diagnosis

Vision foundation models pretrained on large corpora have changed how imaging tasks are approached, replacing task-specific architectures with a large pretrained encoder that is adapted to the target task. Radford et al. [8] introduced CLIP, a contrastive vision-language pretraining approach that aligns image and text representations in a shared embedding space, and this style of pretraining has since been adapted directly to medical imaging. Lin et al. [2] apply the same principle in ADLIP, a vision-language foundation model that jointly encodes MRI scans and structured clinical data for Alzheimer's diagnosis, reporting strong diagnostic performance and even zero-shot prediction ability. This confirms that a vision foundation model, adapted with clinical context, is an effective architecture for this task.

Gap. ADLIP and comparable systems are trained in a centralised setting. They do not address how a vision foundation model should be adapted when the imaging and clinical data it needs are distributed across institutions that cannot share raw data with one another.

2.1.2 Federated Learning for Medical Applications

Federated Learning was formalised by McMahan et al. [9], who proposed FedAvg, in which clients train locally and a central server averages their model updates, weighted by local dataset size, to produce a global model. This formulation underlies almost all subsequent federated medical AI work, including the two studies closest to this project. Mohanraj and Radhakrishnan [1] propose FuzzyFed-CNN, combining MRI-based CNN features with a fuzzy inference layer to produce explainable, federated predictions for early Alzheimer's diagnosis across several simulated hospital clients, and report that the approach can reach high accuracy while preserving privacy. Mouheb et al. [3] benchmark the federated fine-tuning of SAM-Med3D, a pretrained 3D medical foundation model, for dementia classification, comparing classification-head architectures, fine-tuning strategies, and aggregation methods, and find that freezing the foundation model's encoder can match full fine-tuning while advanced aggregation methods outperform plain FedAvg. Parameter-efficient fine-tuning, in particular LoRA [7], is what makes this style of federated foundation-model adaptation practical, since it updates only a small low-rank matrix pair per layer instead of the full parameter set, sharply reducing what each client must transmit.

Gap. Both FuzzyFed-CNN and the federated SAM-Med3D study use standard client participation, either near-uniform participation across simulated clients or standard random sampling, and neither explores how the choice of which clients to aggregate in a given round should depend on the quality or completeness of each client's local data.

2.1.3 Client Selection Strategies in Federated Learning

The general federated learning literature has studied client selection independently of any medical application. Li, Chen, and Teng [4] survey client selection strategies broadly, categorising approaches based on computational resources, data quality, and network conditions, and identify statistical heterogeneity across clients as a central challenge for both convergence speed and fairness. Tang et al. [12] propose FedCor, a correlation-based active client selection strategy that estimates which clients' updates are most likely to reduce the global loss and prioritises them, using a Gaussian-process model of inter-client loss correlation to guide selection without needing labelled validation data from each client.

Gap. These strategies are domain-agnostic. FedCor's correlation model is built from the statistics of model updates, not from any notion of medical data quality, modality completeness, or diagnostic class balance, so it does not by itself capture the constraints that are specific to a federation of hospitals holding multimodal Alzheimer's-diagnosis data.

2.1.4 Explainability in Medical AI

Because a clinician acting on a model's output needs to know why the model produced it, explainability is treated in this project as a first-class requirement rather than an afterthought. Selvaraju et al. [10] introduced Grad-CAM, which uses the gradient of a target class score with respect to a convolutional or attention feature map to produce a coarse localisation of the image regions that most influenced a prediction, and it remains a standard tool for explaining image-based medical predictions. Lundberg and Lee [11] introduced SHAP, a unified, game-theoretically grounded framework for attributing a model's prediction to its input features, which is directly applicable to the structured clinical and demographic variables used in this project. Mohanraj and Radhakrishnan [1] already demonstrate the value of an interpretable, fuzzy-inference layer for Alzheimer's diagnosis, reinforcing that explainability and diagnostic performance are not in conflict for this task.

Gap. None of the reviewed federated Alzheimer's-diagnosis systems pairs image-level explanation (Grad-CAM) with clinical-feature-level explanation (SHAP) within a single federated, multimodal pipeline, and none connects the explanation to which clients contributed most to a given prediction.

2.2 Problem Analysis

The four gaps identified above converge on a single structural problem. Table 2.1 contrasts the closest prior systems with the requirements of this project.

Table 2.1: Contrast between the closest prior work and the proposed framework.

System	Modality	Client Selection	Foundation Model	Explainability
FuzzyFed-CNN [1]	MRI (fuzzy fusion)	Near-uniform	No (CNN)	Fuzzy rules
ADLIP [2]	MRI + clinical text	N/A (centralised)	Yes (vision-language)	Limited
Federated SAM-Med3D [3]	MRI	Random/full	Yes (SAM-Med3D)	No
FedCor [12]	Generic (non-medical)	Correlation-based	No	No
Proposed	MRI + clinical	Adaptive, quality-aware	Yes (ViT-based)	Grad-CAM + SHAP

Three observations follow from Table 2.1. First, every prior system occupies a single cell of the modality-and-selection space: ADLIP has the multimodal foundation model but not the federated setting; FuzzyFed-CNN and the federated SAM-Med3D study have the federated, medical setting but only naive client participation; FedCor has principled client selection but neither the medical domain nor the multimodal foundation model. Second, none of the reviewed federated medical systems reports an explainability component that spans both imaging and clinical modalities. Third, no system in this comparison combines all three requirements, a federated multimodal vision foundation model, adaptive quality-aware client selection, and cross-modal explainability, in one framework.

The problem this project addresses is therefore how to train a single multimodal federated model that predicts Alzheimer's disease status from MRI and clinical data, using a client selection mechanism that accounts for data quality, modality completeness, and class balance instead of random or full participation, while keeping the model's decisions interpretable to clinicians through modality-specific explanations. Chapter 3 sets out the framework and methodology that respond to this problem.

Summary

This chapter reviewed the literature across four themes. Vision foundation models have proven effective for Alzheimer's diagnosis but remain largely centralised. Federated learning provides the privacy-preserving mechanism needed to train across hospitals, and has already been combined with Alzheimer's-specific multimodal models and with foundation-model fine-tuning, but always under naive client participation. General federated learning research has produced principled client selection strategies, but none tailored to medical multimodal data. Explainability tools exist for both imaging and structured clinical data individually, but have not been combined within a federated Alzheimer's-diagnosis pipeline. The gap that remains, and that Chapter 3 addresses, is a single federated, multimodal, foundation-model framework with adaptive client selection and cross-modal explainability.

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